International Mathematics Olympiad 1988 - Question 6 Solution

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Saturday 6th May, 2023

Question:

Let a, b be positive integers such that $a^2 + b^2$ is divisible by $ab + 1$.

Prove that $\frac{a^2+b^2}{ab+1}$ is always a square number.

Solution Development:

Generating pairs:

Let

$$f(a, b) = \frac{a^2 + b^2}{ab + 1} \tag{*}$$

Let us start by finding some pairs of integers such that $f(a, b) \in \mathbb{N}$. Here are all such pairs (a, b) where $a, b < 1000$:

(2, 8), (3, 27), (4, 64), (5, 125), (6, 216), (7, 343), (8, 2), (8, 30), (8, 512), (9, 729), (27, 3), (27, 240), (30, 8), (30, 112), (64, 4), (112, 30), (112, 418), (125, 5), (216, 6), (240, 27), (343, 7), (418, 112), (512, 8), (729, 9)

These pairs were found exhaustively using a Python script. This process has a time complexity of $O(n^2)$ on average and there appears to be an infinite number of solutions, so a proof by exhaustion would be impossible.

Investigating Pairs:

Since addition and multiplication are commutative, we know that if (a, b) is a pair, then (b, a) is also a pair, so we will assume that $a < b$. Notice that there multiple different pairs such that $b = a^3$. Plugging in a general pair of the form (a, a^3) into the function yields

$$f(a, a^3) = \frac{a^2 + (a^3)^2}{a(a^3) + 1} = \frac{a^2 + a^6}{a^4 + 1} = \frac{a^2(1 + a^4)}{1 + a^4} = a^2$$

which is indeed a perfect square.

$$f(a, a^3) = a^2 \tag{**}$$

Now we have shown that there are an infinite number of solutions, but as shown above, not all solutions are of this form. For example,

$$f(8, 30) = \frac{8^2 + 30^2}{8 \times 30 + 1} = \frac{964}{241} = 4$$

If we just look at some more generated pairs that aren't of the form (a, a^3):

(8, 30), (27, 240), (30, 112), (64, 1020), (112, 418), (125, 3120), (216, 7770), (240, 2133), (418, 1560), (1560, 5822)

We see that there appears to be a pattern where the numbers are repeating twice. In fact, if we compute $f(a, b)$ for each pair with a number in common, it is the same result every time. As an example:

$$\begin{cases} f(8, 30) = \frac{8^2+30^2}{8\times30+1} = 4 \\ f(30, 112) = \frac{30^2+112^2}{30\times112+1} = 4 \\ f(112, 418) = \frac{112^2+418^2}{112\times418+1} = 4 \\ \vdots \end{cases}$$

Or, starting with a different pair:

$$\begin{cases} f(27,240) = \frac{27^2+240^2}{27\times 240+1} = 9 \\ f(240,2133) = \frac{240^2+2133^2}{240\times 2133+1} = 9 \\ \vdots \end{cases}$$

Also notice that the pattern $(8,30)$, $(30,112)$, $(112,418)$ starts with $(2,8)$, as $f(2,8) = 4$ as well, while $(27,240)$, $(240,2133)$ starts with $(3,27)$ where the constant function output is $f(3,27) = 9$.

It appears that $f(g,g^3)$, $g \in \mathbb{N}$ generates an infinite number of subsequent solution pairs (a_g, b_g), all with the same common function output of $f(a,b) = g^2$.

Proof:

Firstly, the case of $a = b$ gives

$$f(a,a) = \frac{a^2+a^2}{aa+1} = \frac{2a^2}{a^2+1}$$

For this to be an integer, we need

$$2a^2 = m(a^2+1)$$

$$a^2 = \frac{1}{2-m}$$

where $m \in \mathbb{N}$. Since $a^2 \in \mathbb{N}$, we must have that

$$a = b = m = 1$$

Therefore, the case of $a = b$ results in a perfect square.

We can now assume, without loss of generality, that $a < b$. Using the insights above, the branches of pairs can be described using a system of recurrence relations

$$\begin{cases} a_n = b_{n-1}, & a_1 = g \\ b_n = ma_n - a_{n-1}, & b_1 = g^3 \end{cases} \qquad (***)$$

where g and m are constants, $g, m, n \in \mathbb{N}$.

We will call g the generator, as this value is used to generate an infinite number of pairs.

Lemma 1

$f(a_n, b_n) = f(a_{n+1}, b_{n+1}) = m, \ \ \forall n \in \mathbb{N}$

Proof.

$$f(a_n,b_n) = \frac{a_n^2+b_n^2}{a_nb_n+1} = m \implies a_n^2+b_n^2 = m(a_nb_n+1)$$

$$f(a_{n+1},b_{n+1}) = \frac{b_n^2+(ma_{n+1}-a_n)^2}{(b_n)(ma_{n+1}-a_n)+1} = \frac{b_n^2+m^2a_{n+1}^2-2ma_na_{n+1}+a_n^2}{mb_na_{n+1}-a_nb_n}$$

$$\implies f(a_{n+1},b_{n+1}) = \frac{m^2a_{n+1}^2-2mb_na_n+m(a_nb_n+1)}{mb_na_{n+1}-a_nb_n+1}$$

$$\implies f(a_{n+1},b_{n+1}) = \frac{m(mb_na_{n+1}-a_nb_n+1)}{mb_na_{n+1}-a_nb_n+1} = m$$

□

Lemma 2

$a_n < a_{n+1}$ and $b_n < b_{n+1}$.

Proof. Let (a_n, b_n) be a pair such that $f(a_n, b_n) = m \in \mathbb{N}$, and define

$$b_n = ka_n \iff k = \frac{b_n}{a_n}, \ \ k \in \mathbb{R}$$

$$\implies a_n^2 + (ka_n)^2 = m(a_n(ka_n)+1)$$

$$\implies a_n^2(1-km+k^2) = m$$

Comparing the two sides of the equation, we see that $1 - km + k^2 > 0$

$$\implies k < \frac{m-\sqrt{m^2-4}}{2}, \text{ or } k > \frac{m+\sqrt{m^2-4}}{2}$$

By the triangle inequality, $\sqrt{m^2-4} > m-2$, therefore $k > m-1$.

Expressing a_n and b_n in terms of a_{n+1} and b_{n+1}

$$\begin{aligned} a_n &= ma_{n+1} - b_{n+1} \\ b_n &= a_{n+1} \end{aligned}$$

Since $b_{n+1} = ka_{n+1}$,

$$\begin{aligned} &\implies b_{n+1} > (m-1)a_{n+1} \\ &\implies ma_{n+1} - b_{n+1} < ma_{n+1} - (m-1)a_{n+1} \\ &\implies a_n < ma_{n+1} - ma_{n+1} + a_{n+1} \\ &\implies a_n < a_{n+1} \\ &\implies b_n < b_{n+1} \end{aligned}$$

□

Main Statement

Using $(**)$ and lemma 1, it follows by induction that all pairs (a_n, b_n) generated by some $g \in \mathbb{N}$ are such that $f(a_n, b_n) = g^2$. Therefore, if we can prove that all (a, b) satisfying $(*)$ are generated using $(***)$, then the original statement must be true.

We know from $(***)$ and lemmas 1 and 2 that for any pair (a, b) that satisfy $(*)$ there exists another pair (a', b') with the same function output, and where $a' < a$ and $b' < b$. This process of finding a smaller pair can be repeated an arbitrary number of times and eventually a will be such that $a > 0$ and $a' \leq 0$.

When $a' < 0 \implies b' > 0$ since b' is defined as the previous value of a from $(***)$.

Since a' and b' are integers, $a'b' \leq -1$. If $a'b' = -1$, $f(a', b')$ is undefined, and if $a'b' < -1$, $f(a', b') < 0$ so clearly cannot be equal to m.

Therefore, $a' = 0$

$$\begin{aligned} &\implies ma = b \\ &\implies f(a, ma) = \frac{a^2 + m^2a^2}{ma^2 + 1} = m \\ &\implies a^2 + m^2a^2 = m^2a^2 + m \\ &\implies a^2 = m, \;\; b = a^3 \end{aligned}$$

Therefore, every solution must be either of the form (g, g^3) or must have been generated from a pair of the form (g, g^3), giving a function output of g^2.

$$\therefore \forall (a, b) \in \mathbb{N}^2 \text{ s.t. } a^2 + b^2 \text{ is divisible by } ab + 1, \;\; \frac{a^2 + b^2}{ab + 1} = g^2, \;\; g \in \mathbb{N}. \quad \square$$